

FEATURES

- Member of the Texas Instruments Widebus™ Family
- DOC™ (Dynamic Output Control) Circuit Dynamically Changes Output Impedance, Resulting in Noise Reduction Without Speed Degradation
- Less Than 2-ns Maximum Propagation Delay at 2.5-V and 3.3-V V_{CC}
- Dynamic Drive Capability Is Equivalent to Standard Outputs With I_{OH} and I_{OL} of ± 24 mA at 2.5-V V_{CC}

- Overvoltage-Tolerant Inputs/Outputs Allow Mixed-Voltage-Mode Data Communications
- I_{off} Supports Partial-Power-Down Mode Operation
- Latch-Up Performance Exceeds 100 mA Per JESD 78, Class II
- ESD Protection Exceeds JESD 22
 - 2000-V Human-Body Model (A114-A)
 - 200-V Machine Model (A115-A)

DESCRIPTION/ORDERING INFORMATION

A Dynamic Output Control (DOC) circuit is implemented, which, during the transition, initially lowers the output impedance to effectively drive the load and, subsequently, raises the impedance to reduce noise. Figure 1 shows typical V_{OL} vs I_{OL} and V_{OH} vs I_{OH} curves to illustrate the output impedance and drive capability of the circuit. At the beginning of the signal transition, the DOC circuit provides a maximum dynamic drive that is equivalent to a high-drive standard-output device. For more information, refer to the TI application reports, *AVC Logic Family Technology and Applications*, literature number SCEA006, and *Dynamic Output Control (DOC™) Circuitry Technology and Applications*, literature number SCEA009.

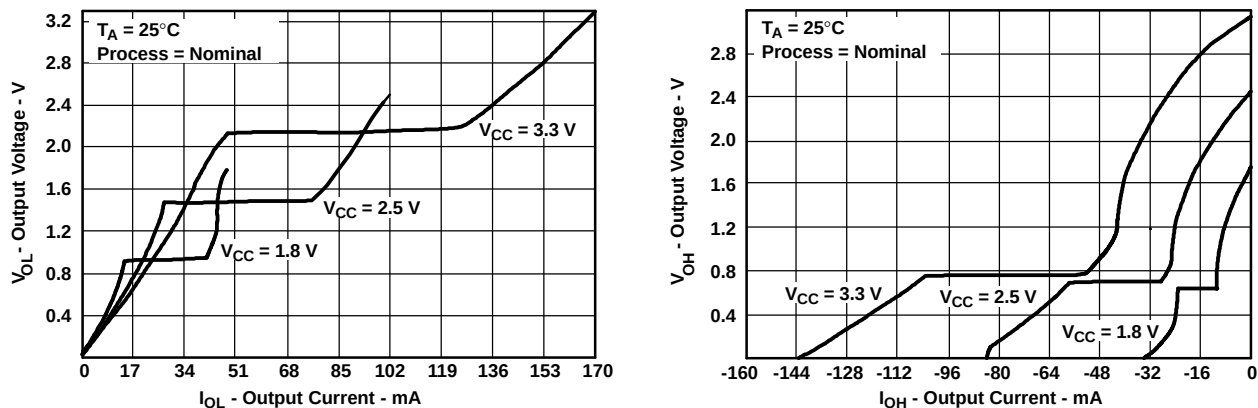


Figure 1. Output Voltage vs Output Current

ORDERING INFORMATION

T_A	PACKAGE ⁽¹⁾		ORDERABLE PART NUMBER	TOP-SIDE MARKING
–40°C to 85°C	TSSOP – DGG	Tape and reel	SN74AVC16244DGGR	AVC16244
	TVSOP – DGV	Tape and reel	SN74AVC16244DGVR	CVA244
	VFBGA – GQL	Tape and reel	SN74AVC16244GQLR	CVA244
	VFBGA – ZQL (Pb-free)		SN74AVC16244ZQLR	

(1) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.



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SN74AVC16244

16-BIT BUFFER/DRIVER

WITH 3-STATE OUTPUTS

SCES141N—JULY 1998—REVISED MARCH 2005

DESCRIPTION/ORDERING INFORMATION (CONTINUED)

This 16-bit buffer/driver is operational at 1.2-V to 3.6-V V_{CC} , but is designed specifically for 1.65-V to 3.6-V V_{CC} operation.

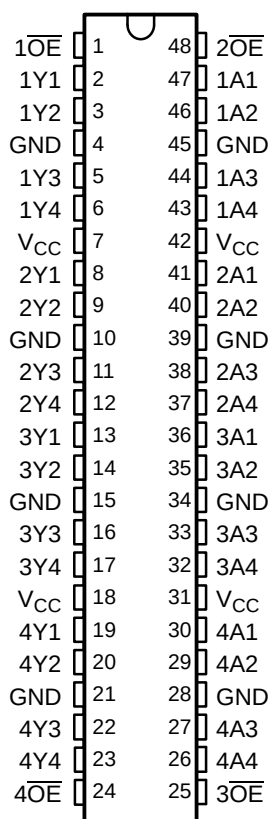
The SN74AVC16244 is designed specifically to improve the performance and density of 3-state memory address drivers, clock drivers, and bus-oriented receivers and transmitters.

The device can be used as four 4-bit buffers, two 8-bit buffers, or one 16-bit buffer. It provides true outputs and symmetrical active-low output-enable ($\overline{OE}$) inputs.

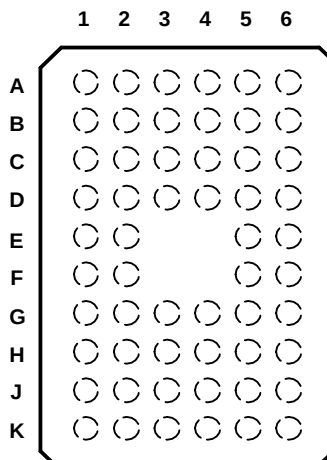
This device is fully specified for partial-power-down applications using I_{off} . The I_{off} circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to V_{CC} through a pullup resistor; the minimum value of the resistor is determined by the current-sinking capability of the driver.

DGG OR DGV PACKAGE
(TOP VIEW)



GQL OR ZQL PACKAGE
(TOP VIEW)



TERMINAL ASSIGNMENTS⁽¹⁾

	1	2	3	4	5	6
A	1 $\overline{OE}$	NC	NC	NC	NC	2 $\overline{OE}$
B	1Y2	1Y1	GND	GND	1A1	1A2
C	1Y4	1Y3	V _{CC}	V _{CC}	1A3	1A4
D	2Y2	2Y1	GND	GND	2A1	2A2
E	2Y4	2Y3			2A3	2A4
F	3Y1	3Y2			3A2	3A1
G	3Y3	3Y4	GND	GND	3A4	3A3
H	4Y1	4Y2	V _{CC}	V _{CC}	4A2	4A1
J	4Y3	4Y4	GND	GND	4A4	4A3
K	4 $\overline{OE}$	NC	NC	NC	NC	3 $\overline{OE}$

(1) NC - No internal connection

FUNCTION TABLE
(EACH 4-BIT BUFFER)

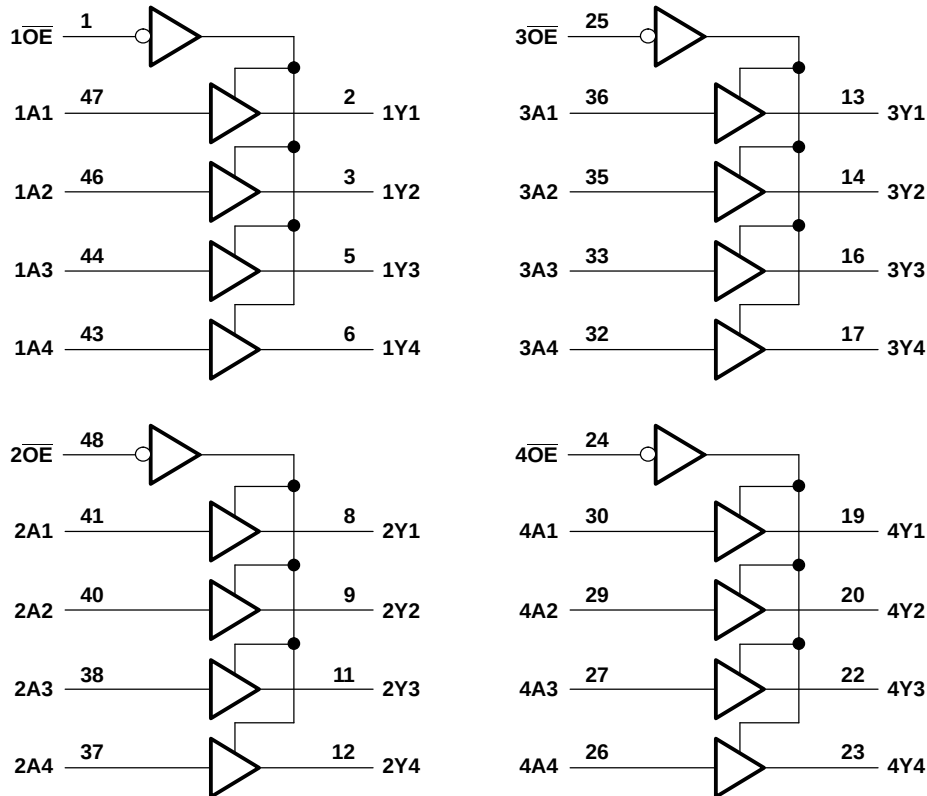
INPUTS		OUTPUT Y
$\overline{OE}$	A	
L	L	L
L	H	H
H	X	Z

SN74AVC16244

16-BIT BUFFER/DRIVER WITH 3-STATE OUTPUTS

SCES141N–JULY 1998–REVISED MARCH 2005

LOGIC DIAGRAM (POSITIVE LOGIC)



Absolute Maximum Ratings⁽¹⁾

over operating free-air temperature range (unless otherwise noted)

			MIN	MAX	UNIT
V _{CC}	Supply voltage range		−0.5	4.6	V
V _I	Input voltage range ⁽²⁾		−0.5	4.6	V
V _O	Voltage range applied to any output in the high-impedance or power-off state ⁽²⁾		−0.5	4.6	V
V _O	Voltage range applied to any output in the high or low state ⁽²⁾⁽³⁾		−0.5	V _{CC} + 0.5	V
I _{IK}	Input clamp current	V _I < 0	−50		mA
I _{OK}	Output clamp current	V _O < 0	−50		mA
I _O	Continuous output current		±50		mA
Continuous current through each V _{CC} or GND			±100		mA
θ _{JA}	Package thermal impedance ⁽⁴⁾	DGG package	70		°C/W
		DGV package	58		
		GQL/ZQL package	42		
T _{stg}	Storage temperature range		−65	150	°C

- (1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The input and output negative-voltage ratings may be exceeded if the input and output current ratings are observed.
- (3) The output positive-voltage rating may be exceeded up to 4.6 V maximum if the output current rating is observed.
- (4) The package thermal impedance is calculated in accordance with JESD 51.

Recommended Operating Conditions⁽¹⁾

		MIN	MAX	UNIT
V _{CC}	Supply voltage	Operating	1.4	3.6
		Data retention only	1.2	
V _{IH}	High-level input voltage	V _{CC} = 1.2 V	V _{CC}	V
		V _{CC} = 1.4 V to 1.6 V	0.65 × V _{CC}	
		V _{CC} = 1.65 V to 1.95 V	0.65 × V _{CC}	
		V _{CC} = 2.3 V to 2.7 V	1.7	
		V _{CC} = 3 V to 3.6 V	2	
V _{IL}	Low-level input voltage	V _{CC} = 1.2 V	GND	V
		V _{CC} = 1.4 V to 1.6 V	0.35 × V _{CC}	
		V _{CC} = 1.65 V to 1.95 V	0.35 × V _{CC}	
		V _{CC} = 2.3 V to 2.7 V	0.7	
		V _{CC} = 3 V to 3.6 V	0.8	
V _I	Input voltage	0	3.6	V
V _O	Output voltage	Active state	0	V _{CC}
		3-state	0	3.6
I _{OHS}	Static high-level output current ⁽²⁾	V _{CC} = 1.4 V to 1.6 V	–2	mA
		V _{CC} = 1.65 V to 1.95 V	–4	
		V _{CC} = 2.3 V to 2.7 V	–8	
		V _{CC} = 3 V to 3.6 V	–12	
I _{OLS}	Static low-level output current ⁽²⁾	V _{CC} = 1.4 V to 1.6 V	2	mA
		V _{CC} = 1.65 V to 1.95 V	4	
		V _{CC} = 2.3 V to 2.7 V	8	
		V _{CC} = 3 V to 3.6 V	12	
Δt/Δv	Input transition rise or fall rate	V _{CC} = 1.4 V to 3.6 V	5	ns/V
T _A	Operating free-air temperature	–40	85	°C

- (1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.
- (2) Dynamic drive capability is equivalent to standard outputs with I_{OH} and I_{OL} of ±24 mA at 2.5-V V_{CC}. See Figure 1 for V_{OL} vs I_{OL} and V_{OH} vs I_{OH} characteristics. Refer to the TI application reports, *AVC Logic Family Technology and Applications*, literature number SCEA006, and *Dynamic Output Control (DOC™) Circuitry Technology and Applications*, literature number SCEA009.

SN74AVC16244

16-BIT BUFFER/DRIVER

WITH 3-STATE OUTPUTS

SCES141N–JULY 1998–REVISED MARCH 2005



Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	V _{CC}	MIN	TYP ⁽¹⁾	MAX	UNIT
V _{OH}		I _{OHS} = –100 µA	1.4 V to 3.6 V	V _{CC} – 0.2			V
		I _{OHS} = –2 mA, V _{IH} = 0.91 V	1.4 V	1.05			
		I _{OHS} = –4 mA, V _{IH} = 1.07 V	1.65 V	1.2			
		I _{OHS} = –8 mA, V _{IH} = 1.7 V	2.3 V	1.75			
		I _{OHS} = –12 mA, V _{IH} = 2 V	3 V	2.3			
V _{OL}		I _{OLS} = 100 µA	1.4 V to 3.6 V			0.2	V
		I _{OLS} = 2 mA, V _{IL} = 0.49 V	1.4 V			0.4	
		I _{OLS} = 4 mA, V _{IL} = 0.57 V	1.65 V			0.45	
		I _{OLS} = 8 mA, V _{IL} = 0.7 V	2.3 V			0.55	
		I _{OLS} = 12 mA, V _{IL} = 0.8 V	3 V			0.7	
I _I		V _I = V _{CC} or GND	3.6 V			±2.5	µA
I _{off}		V _I or V _O = 3.6 V	0			±10	µA
I _{OZ}		V _O = V _{CC} or GND	3.6 V			±10	µA
I _{CC}		V _I = V _{CC} or GND, I _O = 0	3.6 V			40	µA
C _i	Control inputs	V _I = V _{CC} or GND	2.5 V	3.5			pF
			3.3 V	3.5			
	Data inputs	V _I = V _{CC} or GND	2.5 V	6			
			3.3 V	6			
C _O	Outputs	V _O = V _{CC} or GND	2.5 V	6.5			pF
			3.3 V	6.5			

(1) Typical values are measured at T_A = 25°C.

Switching Characteristics

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 2)

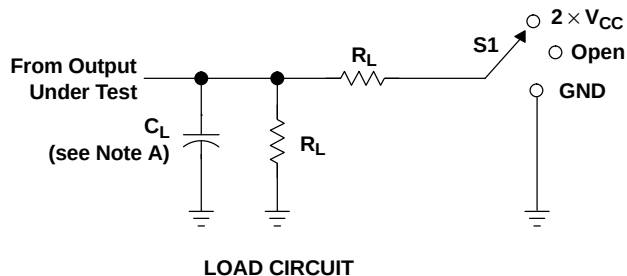
PARAMETER	FROM (INPUT)	TO (OUTPUT)	V _{CC} = 1.2 V	V _{CC} = 1.5 V ± 0.1 V		V _{CC} = 1.8 V ± 0.15 V		V _{CC} = 2.5 V ± 0.2 V		V _{CC} = 3.3 V ± 0.3 V		UNIT
			TYP	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t _{pd}	A	Y	3.1	0.6	3.3	0.7	2.9	0.6	1.9	0.5	1.7	ns
t _{en}	$\overline{OE}$	Y	7.6	1.4	8	1.3	6.8	0.9	4	0.7	3.5	ns
t _{dis}	$\overline{OE}$	Y	7.2	1.7	7.3	1.6	6.2	1	4.3	1	3.5	ns

Operating Characteristics

T_A = 25°C

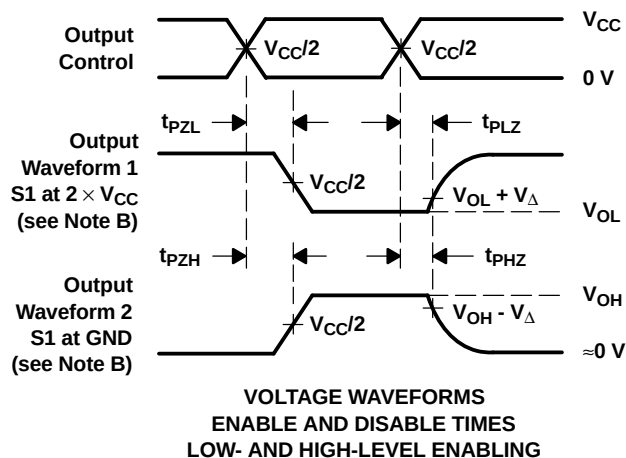
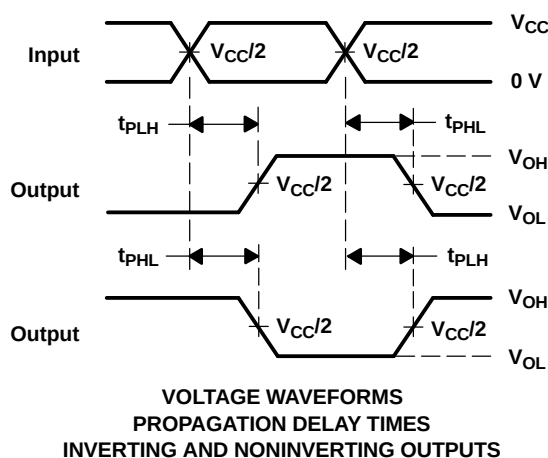
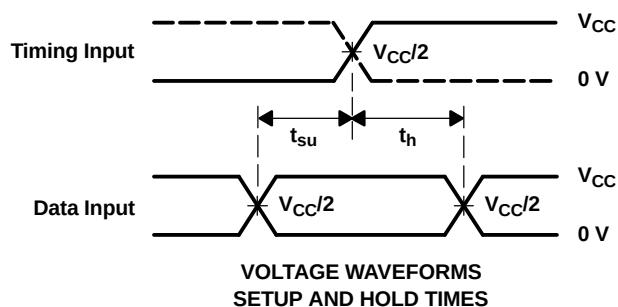
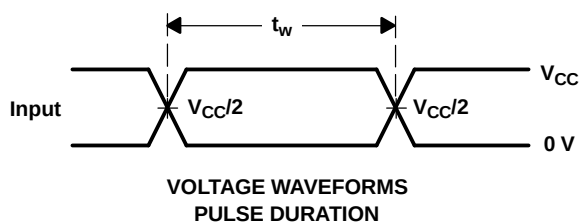
PARAMETER			TEST CONDITIONS	V _{CC} = 1.8 V	V _{CC} = 2.5 V	V _{CC} = 3.3 V	UNIT
				TYP	TYP	TYP	
C _{pd}	Power dissipation capacitance	Outputs enabled	C _L = 0, f = 10 MHz	23	27	33	pF
		Outputs disabled		0.1	0.1	0.1	

PARAMETER MEASUREMENT INFORMATION



TEST	S1
t_{PLH}/t_{PHL}	Open
t_{PLZ}/t_{PZL}	$2 \times V_{CC}$
t_{PHZ}/t_{PZH}	GND

V_{CC}	C_L	R_L	V_{Δ}
1.2 V	15 pF	2 k Ω	0.1 V
1.5 V $\pm$ 0.1 V	15 pF	2 k Ω	0.1 V
1.8 V $\pm$ 0.15 V	30 pF	1 k Ω	0.15 V
2.5 V $\pm$ 0.2 V	30 pF	500 Ω	0.15 V
3.3 V $\pm$ 0.3 V	30 pF	500 Ω	0.3 V



- NOTES:
- C_L includes probe and jig capacitance.
 - Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
 - All input pulses are supplied by generators having the following characteristics: PRR $\leq$ 10 MHz, $Z_O = 50 \Omega$, slew rate $\geq$ 1 V/ns.
 - The outputs are measured one at a time, with one transition per measurement.
 - t_{PLZ} and t_{PHZ} are the same as t_{dis} .
 - t_{PZL} and t_{PZH} are the same as t_{en} .
 - t_{PLH} and t_{PHL} are the same as t_{pd} .
 - All parameters and waveforms are not applicable to all devices.

Figure 2. Load Circuit and Voltage Waveforms

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74AVC16244DGGR	Active	Production	TSSOP (DGG) 48	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	AVC16244
SN74AVC16244DGVR	Active	Production	TVSOP (DGV) 48	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	CVA244

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

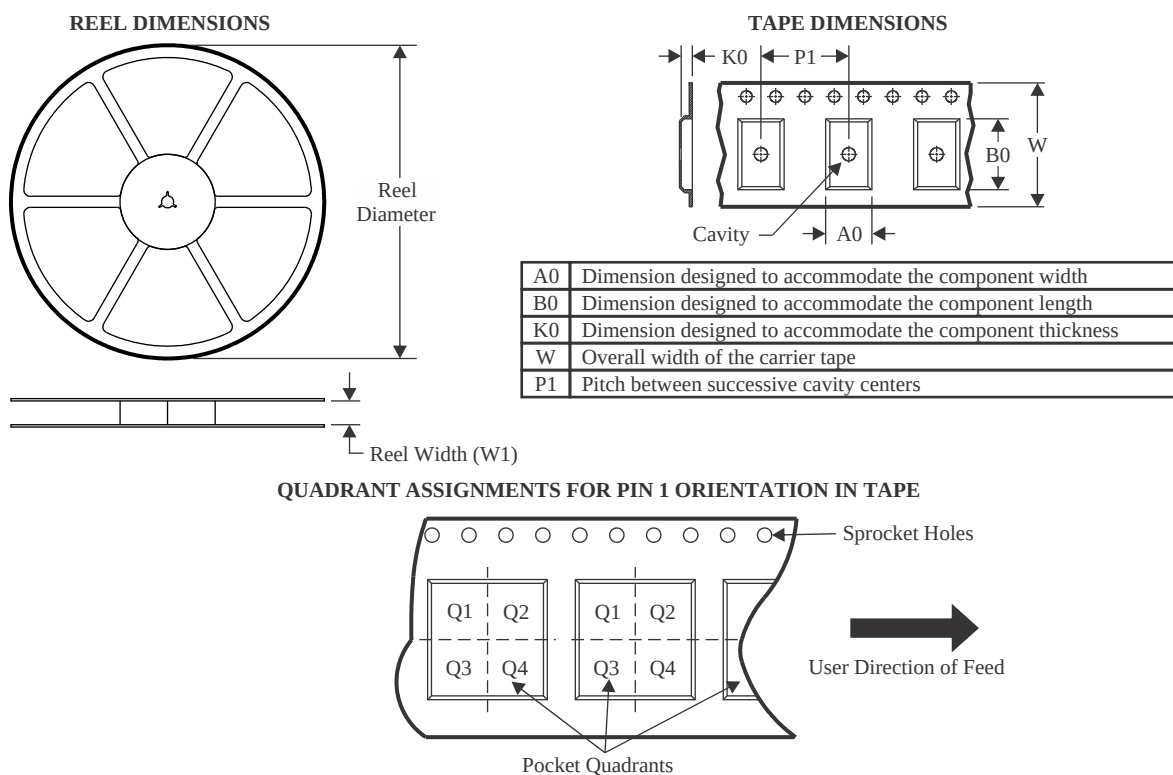
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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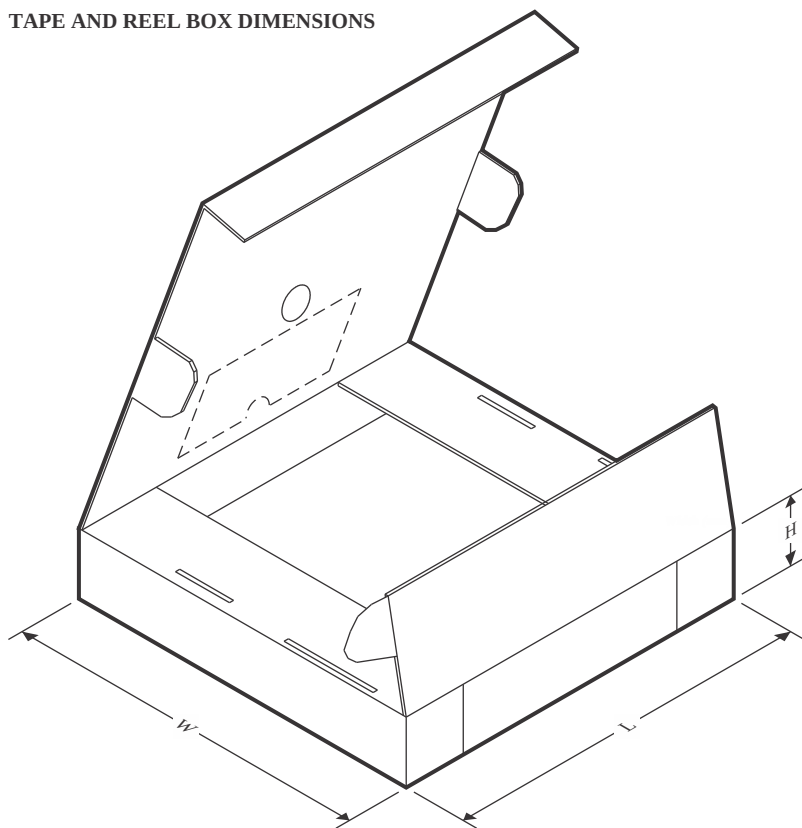
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74AVC16244DGGR	TSSOP	DGG	48	2000	330.0	24.4	8.6	13.0	1.8	12.0	24.0	Q1
SN74AVC16244DGVR	TVSOP	DGV	48	2000	330.0	16.4	7.1	10.2	1.6	12.0	16.0	Q1

TAPE AND REEL BOX DIMENSIONS



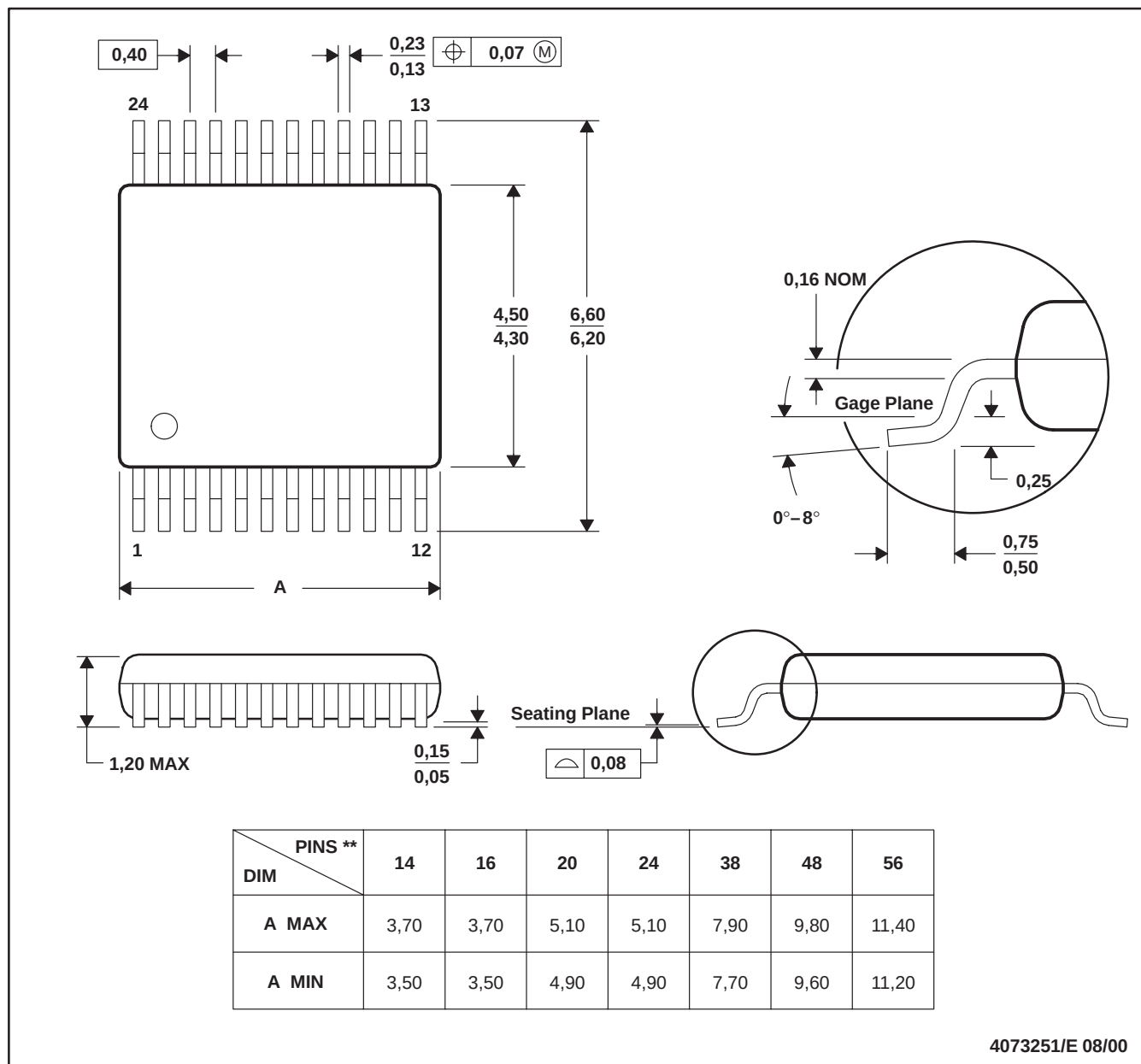
*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74AVC16244DGGR	TSSOP	DGG	48	2000	367.0	367.0	45.0
SN74AVC16244DGVR	TVSOP	DGV	48	2000	356.0	356.0	35.0

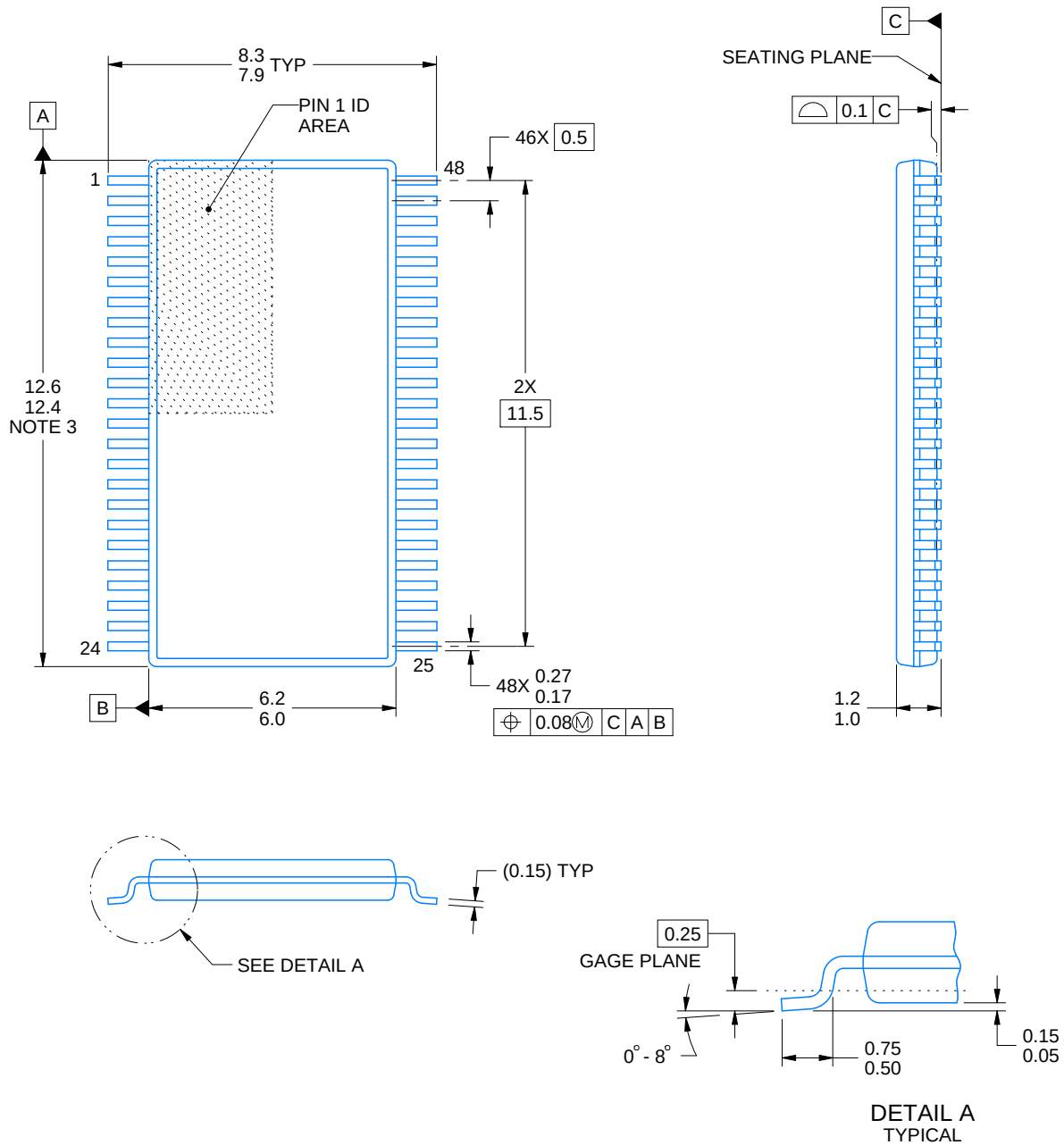
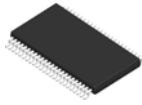
DGV (R-PDSO-G**)

PLASTIC SMALL-OUTLINE

24 PINS SHOWN



- NOTES: A. All linear dimensions are in millimeters.
 B. This drawing is subject to change without notice.
 C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15 per side.
 D. Falls within JEDEC: 24/48 Pins – MO-153
 14/16/20/56 Pins – MO-194



4214859/B 11/2020

NOTES:

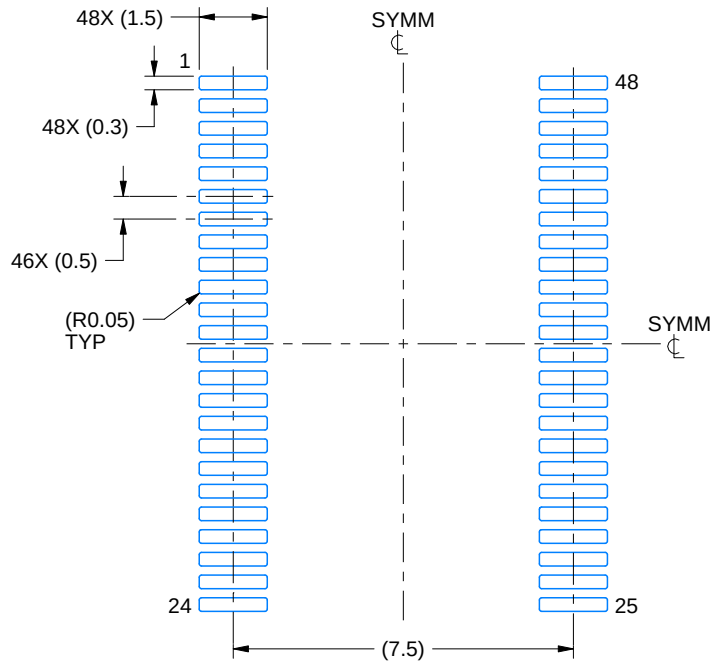
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

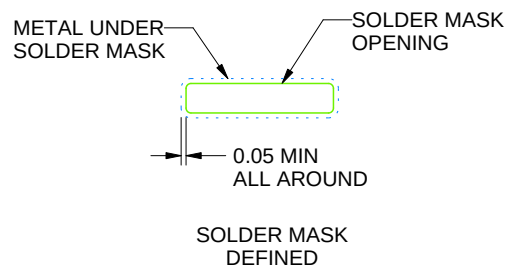
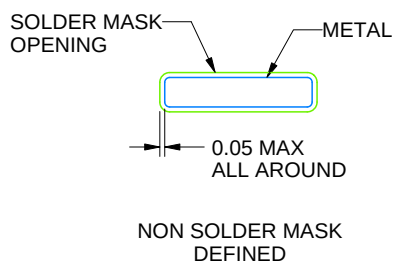
DGG0048A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
SCALE:6X



SOLDER MASK DETAILS

4214859/B 11/2020

NOTES: (continued)

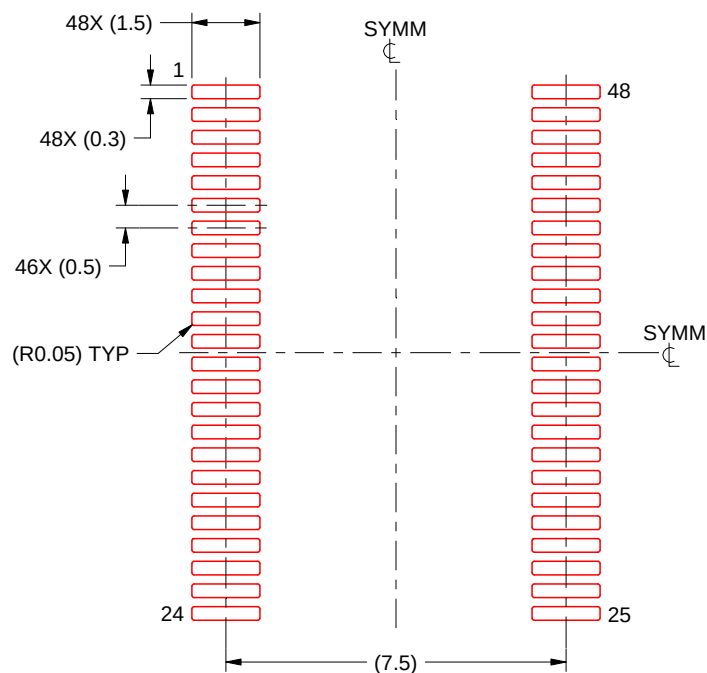
5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DGG0048A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

4214859/B 11/2020

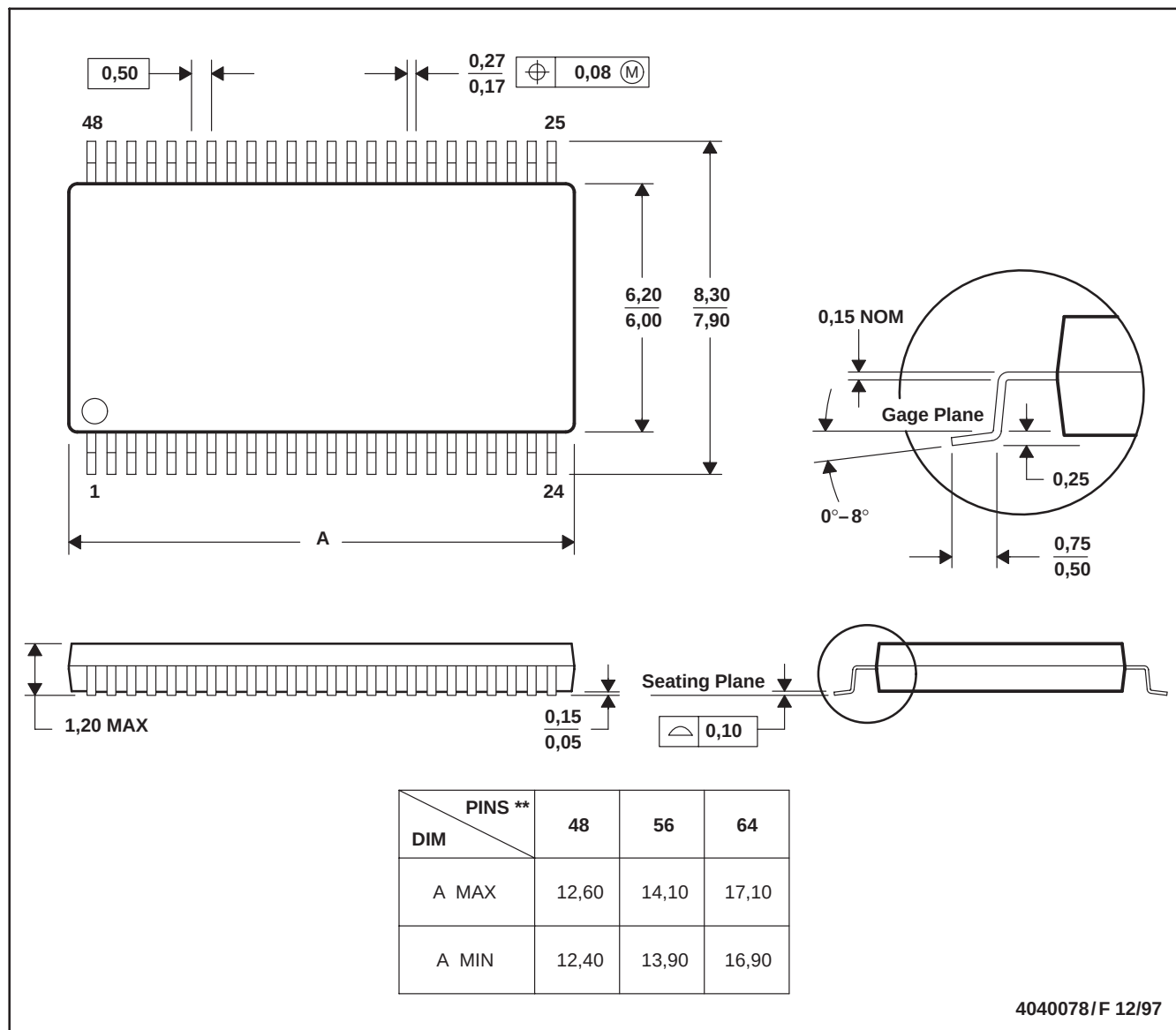
NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

DGG (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

48 PINS SHOWN



- NOTES: A. All linear dimensions are in millimeters.
 B. This drawing is subject to change without notice.
 C. Body dimensions do not include mold protrusion not to exceed 0,15.
 D. Falls within JEDEC MO-153

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